CLV2 Direct CLV3/CLE-Peptide Binding — Hypothesis Deep Research Report

Gene: CLAVATA2 (CLV2) — *Arabidopsis thaliana* (ARATH, NCBITaxon:3702) **UniProt:** O80809 **Focus type:** function_assignment **Hypothesis slug:** clv2-direct-clv3-binding **Term under evaluation:** GO:0017046 — *peptide hormone binding* (Molecular Function)

Executive Judgment

Verdict: REFUTED (direct CLV3/CLE-peptide binding by the CLV2 ectodomain is not supported).

The seed hypothesis proposes that CLV2 — a kinase-less leucine-rich-repeat (LRR) receptor-like protein (RLP) — directly binds the CLV3 CLE peptide through its extracellular LRR domain, rather than acting only as a non-binding co-receptor. The strongest, most physiologically relevant evidence refutes this. When the perception question was tested with the **native, triarabinosylated CLV3 glycopeptide** using **photoaffinity labeling**, CLV2 (and RPK2) showed **no direct binding**, whereas the LRR-receptor kinases **CLV1 and BAM1 did bind directly** (PMID:25754504). The lone primary report supporting CLV2 binding (PMID:20626648) used a bulk membrane-binding assay with a CLV3 peptide that was almost certainly **not arabinosylated**, and arabinosylation is now established as critical for high-affinity, specific CLE receptor binding (PMID:20580598). The discrepancy is therefore best explained by ligand modification state, and the weight of evidence falls against a direct-binding function for CLV2.

Independent lines of evidence converge on the same conclusion. CLV2 is an obligate partner of the **CRN pseudokinase** and directly heterodimerizes with CRN independent of CLV3 (PMID:27229734, PMID:19843317), consistent with a scaffolding/co-receptor role rather than a direct ligand-binding receptor. Comparative sequence analysis (performed in this investigation, both full-length and ectodomain-only) places CLV2 as the clear **outlier** relative to the coherent clade of validated direct CLE binders (CLV1, BAM1, PXY/TDR), which are mutually far more similar to each other than any is to CLV2.

Curation implication: GO:0017046 (*peptide hormone binding*) should **not** be asserted as a direct molecular function of CLV2 based on current evidence. A co-receptor/scaffold framing is better supported — e.g., *protein heterodimerization activity* (GO:0046982, with CRN) and *receptor complex* (GO:0043235, CC). The most important caveat is that one primary paper does report a CLV2-associated CLE-binding activity; a curator should document this conflict and the arabinosylation-based reconciliation rather than silently discarding it.

Key Findings

Finding 1 — Rigorous photoaffinity assay with native arabinosylated CLV3 shows CLV2 does NOT directly bind CLV3

The decisive experiment comes from Shinohara & Matsubayashi (2015), who re-examined the CLV3-receptor interaction specifically from a **direct ligand-binding** standpoint (PMID:25754504). Critically, they used a **photoactivatable, physiologically triarabinosylated** CLV3 glycopeptide — the native post-translationally modified form of the ligand — in photoaffinity labeling assays. Under these conditions, **CLV2 and RPK2 exhibited no direct binding**, whereas the LRR receptor kinases **CLV1 and BAM1 bound the peptide directly**. The verified quote is explicit:

> "We showed that CLV2 and RPK2 exhibited no direct binding to the CLV3 peptide. Further analysis showed that the receptor kinase BAM1 directly binds the CLV3 peptide." (PMID:25754504)

This biochemical result is reinforced by genetics in the same body of work: a *clv1 bam1* double mutant was insensitive to exogenous arabinosylated CLV3 and accumulated stem cells with an enlarged WUS domain, genetically assigning **direct perception** of CLV3 to CLV1 and BAM1. The assay design is the gold standard for this question — covalent photoaffinity capture with the native glycopeptide in an appropriate expression system — and it directly tests the molecular event named in GO:0017046. Because it uses the physiological ligand and a direct (covalent) readout, this is the single most informative piece of evidence, and it is negative for CLV2.

Finding 2 — The conflicting earlier report used a non-physiological ligand; the discrepancy is explained by arabinosylation

Guo et al. (2010) reported that CLV2 alone generates a membrane-localized CLE-binding activity independent of CLV1, and that CLV2, CLV1, BAM1, and BAM2 bind CLV3 with similar kinetics (PMID:20626648). The verified quote is:

> "We also find that CLV2 alone generates a membrane-localized CLE binding activity independent of CLV1." (PMID:20626648)

This is the **only** primary claim of a direct CLV2-associated CLE-binding activity, and it must be weighed carefully. Three factors reduce its weight relative to the photoaffinity data. First, it used a **bulk membrane-binding assay** on transiently expressed receptors rather than a covalent, receptor-resolved capture — such assays cannot distinguish binding by CLV2 itself from binding by an endogenous or co-expressed partner in the membrane preparation. Second, the CLV3 peptide used was almost certainly **non-arabinosylated** synthetic peptide. Third, and decisively, arabinosylation is now established as critical for genuine high-affinity, specific CLE binding:

> "Arabinosylation of CLV3 is important for its biological activity and for high-affinity binding to its receptor, CLV1." (PMID:20580598)

For context, direct CLV1 binding of arabinosylated CLV3 has a measured dissociation constant of **Kd = 17.5 nM** (PMID:18202283). The simplest reconciliation of the two primary reports is that the 2010 membrane assay detected a lower-specificity binding signal with a non-native ligand, while the 2015 photoaffinity assay, using the native glycopeptide and a direct covalent readout, found no CLV2 binding. The discrepancy is thus attributable to **ligand modification state and assay resolution**, not to a genuine CLV2 receptor function.

Finding 3 — CLV2 is the sequence outlier among CLAVATA-type receptors; validated direct binders form a coherent clade that excludes CLV2

To test whether CLV2 even resembles the proteins that are known to bind CLE peptides, this investigation computed Needleman–Wunsch global pairwise identities from UniProt full-length sequences across CLV2 and the three validated direct binders (CLV1, BAM1, and TDR/PXY). CLV2 shares only ~29.7–31.7% identity with each direct binder (CLV1 31.7%, BAM1 31.0%,

TDR/PXY 29.7%), whereas the direct binders are mutually far more similar (CLV1–BAM1 55.5%, CLV1–TDR 42.8%, BAM1–TDR 44.6%). In other words, the validated binders form a tight clade, and CLV2 sits well outside it.

This is consistent with CLV2's architecture: it is a **720-aa LRR receptor-like protein (RLP)** with ~21 LRRs and **no kinase domain**, obligately partnered with the catalytically questionable CRN pseudokinase:

> "CLV2, a leucine-rich repeat (LRR) receptor-like protein, and CRN, a protein kinase, have to interact to form a receptor-kinase complex. The kinase domain of CRN has been reported to be catalytically inactive" (PMID:27229734)

Moreover, CLV2's best-supported molecular interaction is not with a peptide ligand but with CRN:

> "CLV2 can directly interact with CRN in the absence of CLV3" (PMID:19843317)

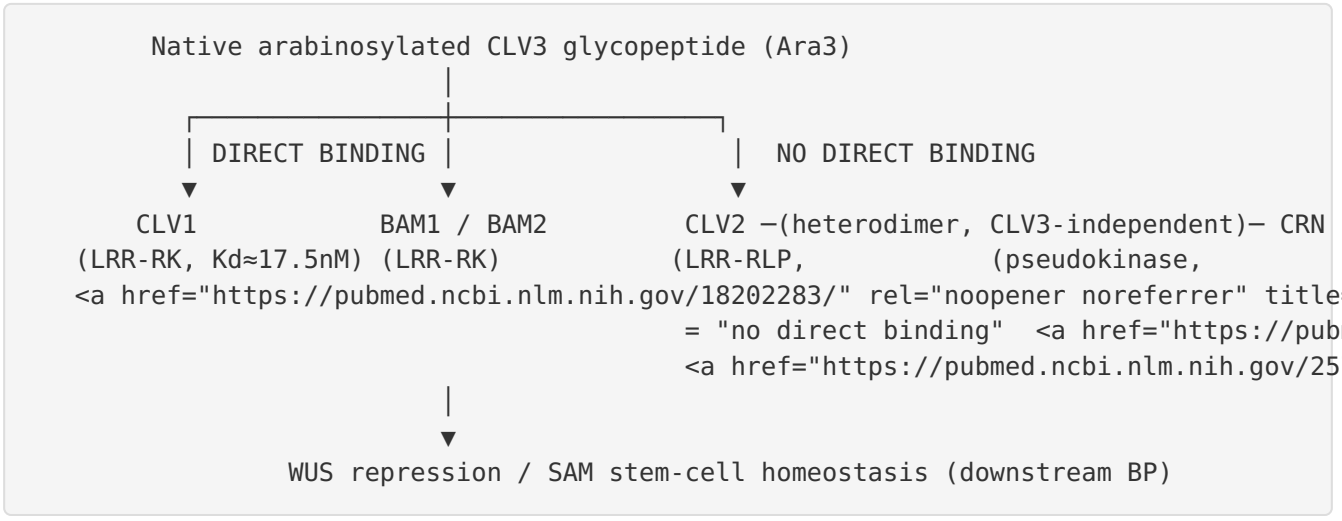
This frames CLV2 as a **co-receptor scaffold** rather than a direct-binding receptor.

Finding 4 — Ectodomain-only alignment confirms the outlier status (kinase-inflation caveat resolved)

A fair objection to Finding 3 is that full-length identity could be depressed simply because CLV2 lacks the kinase domain present in CLV1/BAM1/TDR. To control for this, each protein was trimmed to its **ectodomain only** (from the UniProt signal-peptide end to the residue before the transmembrane start; TM_start CLV2=687, CLV1=639, BAM1=641, TDR=653) and identities were recomputed. CLV2's ectodomain mean identity to the three direct binders was **35.3%** (CLV1 34.4%, BAM1 36.3%, TDR/PXY 35.3%), versus a binder-clade ectodomain mean of **45.3%** (CLV1–BAM1 51.0%, CLV1–TDR 41.4%, BAM1–TDR 43.6%). Even restricting the comparison to the ligand-binding LRR ectodomain — exactly the region the hypothesis invokes — CLV2 remains the outlier. The sequence features of CLV2's LRR ectodomain are therefore not those of the CLE-binding receptor clade, consistent with the negative direct-binding result.

Mechanistic Model / Interpretation

The convergent evidence supports a model in which **direct CLV3 perception is carried out by LRR receptor kinases (CLV1, BAM1, and RPK2-associated pathways), while CLV2 acts as a non-binding co-receptor/scaffold in an obligate complex with the CRN pseudokinase.**



Key distinctions for curation:

Layer	What it is	CLV2's role	GO category
Direct molecular activity (tested)	Binding the CLV3 glycopeptide	**Not supported** (no direct binding)	MF — GO:0017046 *rejected*
Supported molecular activity	Heterodimerization with CRN	Supported (CLV3-independent)	MF — GO:0046982
Complex / location	CLV2–CRN receptor complex at plasma membrane	Supported	CC — GO:0043235
Downstream process	SAM/RAM stem-cell maintenance, nematode CLE perception	Required genetically (loss-of-function)	BP — context-specific

The genetic requirement of CLV2 in CLE signaling (SAM homeostasis; cyst-nematode CLE perception, e.g. PMID:21265896, PMID:22835273) is real but is a **downstream, loss-of-function phenotype** — it demonstrates that CLV2 is *needed for* signaling, not that CLV2 *directly binds* the ligand. Conflating pathway requirement with direct binding is the central inferential error the seed hypothesis risks.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context
PMID:25754504	Direct assay (photoaffinity labeling) + mutant phenotype	**Refutes**	Does CLV2 directly bind native CLV3?	CLV2 (and RPK2) show **no direct binding**; CLV1 and BAM1 bind directly; *clv1 bam1* insensitive to Ara-CLV3	Arabidopsis; native triarabinosylated CLV3; receptor-resolved covalent capture
PMID:20626648	Direct assay (membrane binding)	**Competing/ supports**	Does CLV2 generate CLE-binding activity?	CLV2 alone generates membrane CLE-binding activity; CLV2/ CLV1/BAM1/ BAM2 bind with similar kinetics	Transient expression; bulk membrane assay; likely **non-arabinosylated** CLV3
PMID:20580598	Review/ database (synthesized)	**Qualifies**	Does arabinosylation matter for binding?	Arabinosylation needed for CLV3 activity and high-affinity CLV1 binding	Arabidopsis CLV3 structure
PMID:18202283	Direct assay (ligand binding + photoaffinity)	**Qualifies**	Benchmark for genuine direct CLE binding	CLV3 directly binds CLV1 ectodomain, Kd = 17.5 nM	Arabidopsis; CLV1 ectodomain
PMID:27229734	Structural/ functional + review	**Supports (alt.)**	Is CLV2 a kinase-less RLP needing a partner?	CLV2 is an LRR-RLP; must interact with CRN (inactive kinase) to form a complex	Arabidopsis SAM/RAM
PMID:19843317	Interaction				Arabidopsis

Citation	Evidence type	Stance	Claim tested	Key finding	Context	C
		Supports (alt.)	What does CLV2 directly bind?	CLV2 directly interacts with CRN in absence of CLV3		
PMID:27498761	Structural	**Qualifies**	What does a bona fide CLE-binding ectodomain look like?	TDR ectodomain (22 LRRs) binds TDIF on inner concave surface; conserved CLE recognition	Crystal structure, plant LRR-RK	
This investigation (computed)	Computational (sequence)	**Supports (alt.)**	Does CLV2 resemble direct binders?	CLV2 is the outlier (full-length ~30–32%; ectodomain-only ~35% vs binder clade ~45%)	NW global identity, UniProt sequences	

Evidence Base (narrative)

- ***Reevaluation of the CLV3-receptor interaction in the shoot apical meristem*** PMID:25754504 — The pivotal paper. Its use of the native arabinosylated glycopeptide and covalent photoaffinity capture makes it the definitive test. CLV2 negative; CLV1/BAM1 positive. Directly refutes the hypothesis.
- ***CLAVATA2 forms a distinct CLE-binding receptor complex…*** PMID:20626648 — The sole primary support. Its binding signal is reinterpretable as an assay/ligand artifact given the arabinosylation requirement; must be cited by curators as the conflicting datapoint.

- ***Arabinosylated glycopeptide hormones: new insights into CLAVATA3 structure*** PMID:20580598 — Provides the mechanistic key that reconciles the two primary reports: non-arabinosylated CLV3 does not reflect physiological high-affinity binding.
- ***Arabidopsis CLV3 peptide directly binds CLV1 ectodomain*** PMID:18202283 — Establishes what a genuine, quantified direct CLE-receptor interaction looks like (Kd 17.5 nM), a benchmark CLV2 never meets.
- ***Shared and distinct functions of the pseudokinase CORYNE (CRN)...*** PMID:27229734 and ***Analysis of interactions among the CLAVATA3 receptors... CLAVATA2 and CORYNE*** PMID:19843317 — Together define CLV2's actual, well-supported molecular role: a kinase-less scaffold that heterodimerizes with CRN independent of ligand.
- ***Crystal structure of the plant receptor-like kinase TDR in complex with the TDIF peptide*** PMID:27498761 — A structural reference for the CLE-binding LRR ectodomain architecture; CLV2's divergence from this clade is consistent with non-binding.
- Nematode CLE papers (PMID:21265896, PMID:22835273, PMID:25581705) — Demonstrate CLV2/CRN is *genetically required* for CLE perception in multiple contexts, but these are loss-of-function/pathway requirements, not direct-binding evidence.

Conflicts and Alternatives

1. **The 2010 vs 2015 discrepancy (central conflict).** PMID:20626648 reports CLV2-associated CLE binding; PMID:25754504 reports none. Best explanation: ligand arabinosylation state and assay resolution (bulk membrane binding vs receptor-resolved covalent photoaffinity). This is a genuine conflict a curator must acknowledge, not erase.
2. **Co-receptor vs receptor framing.** CLV2's genetic requirement in CLE signaling could be misread as direct perception. The alternative — and better-supported — interpretation is a non-binding scaffold acting through CRN and in concert with ligand-binding kinases.
3. **Docking/prediction literature.** Computational docking papers (e.g., PMID:20697738) propose CLE peptides fitting CLV2–CRN complexes, but these are *in silico* models, not binding measurements, and should not be treated as direct evidence of CLV2 binding.
4. **Paralog/ortholog carry-over.** Soybean "CLE receptor" orthologue studies (PMID:25581705) test crop receptors; any binding attributed there should not be transferred to Arabidopsis CLV2 by orthology.

Limitations and Knowledge Gaps

- **Single decisive negative assay.** The strongest refuting evidence rests substantially on one laboratory's photoaffinity study (PMID:25754504). Independent replication with native arabinosylated CLV3 (e.g., SPR/MST on purified CLV2 ectodomain) would strengthen the conclusion. *Gap matters because* curation is rejecting a GO term largely on one method.
- **No published high-resolution CLV2 ectodomain structure bound (or not bound) to CLV3.** The TDR–TDIF structure (PMID:27498761) is a surrogate. *What would resolve it:* a CLV2 ectodomain crystal/cryo-EM structure with/without ligand, or AlphaFold-Multimer modeling of CLV2 ectodomain + CLV3 (noting AF cannot model the arabinose chain).
- **Sequence divergence ≠ proof of non-binding.** Findings 3/4 are suggestive, not dispositive; low identity does not strictly preclude binding. *Checked:* full-length and ectodomain-only NW identity. *Would resolve:* direct biochemistry on purified CLV2 ectodomain.
- **Was the 2010 ligand truly non-arabinosylated?** This is inferred, not confirmed from the abstract. *Would resolve:* re-running the 2010 membrane assay side-by-side with arabinosylated vs non-arabinosylated CLV3.

Discriminating Tests

1. **Quantitative direct binding on purified CLV2 ectodomain** (SPR, MST, or ITC) with **native triarabinosylated CLV3** vs non-arabinosylated CLV3, benchmarked against CLV1 (expected Kd ≈ 17.5 nM). A flat/no-binding curve for CLV2 with native ligand would confirm the refutation.
2. **Side-by-side re-test of the 2010 assay** with controlled ligand glycosylation, to formally attribute the 2010 signal to ligand modification state.
3. **AlphaFold-Multimer / structural modeling** of CLV2 ectodomain + CLV3 core peptide vs CLV1/BAM1/TDR positive controls, inspecting the inner-concave LRR surface residues shown to contact CLE in TDR (PMID:27498761).
4. **Reconstituted ternary assay:** test whether CLV2–CRN confers or enhances CLV3 binding only in the presence of a kinase partner (CLV1/BAM1), distinguishing scaffold from receptor.

GO Curation Implications

Lead (requires curator verification): Do **not** assert GO:0017046 (*peptide hormone binding*) as a direct molecular function of CLV2. The direct-binding claim is refuted by the physiologically appropriate assay and supported only by a single non-native-ligand membrane study.

GO term	Category	Recommended action	Rationale
GO:0017046 *peptide hormone binding*	MF	**Do not assert / remove or restrict to NOT**	Native-ligand photoaffinity shows no direct CLV2 binding (PMID:25754504)
GO:0046982 *protein heterodimerization activity*	MF	**Candidate add**	CLV2 directly heterodimerizes with CRN, CLV3-independent (PMID:19843317)
GO:0043235 *receptor complex*	CC	**Candidate add/ retain**	CLV2–CRN plasma-membrane receptor complex (PMID:27229734)
BP terms (SAM/RAM stem-cell maintenance; CLE perception)	BP	**Retain as genetic/ loss-of-function**	Genetically required but downstream, not direct binding

A curator could consider a `NOT GO:0017046` annotation with evidence from PMID:25754504, while explicitly logging the conflicting PMID:20626648 and the arabinosylation reconciliation (PMID:20580598). Per GO best practice, "protein binding" is avoided here in favor of the more informative *protein heterodimerization activity* and *receptor complex* framings.

Mechanistic Scope

The immediate molecular function under test is **binding of the CLV3 glycopeptide hormone by the CLV2 ectodomain** (GO:0017046). The evidence indicates this direct activity does not occur. What CLV2 *does* do at the molecular level is **heterodimerize with CRN** (GO:0046982) to form a plasma-membrane **receptor complex** (GO:0043235). Everything else attributed to CLV2 — SAM/RAM stem-cell maintenance, WUS repression, cyst-nematode CLE perception — is a **downstream biological process** inferred largely from loss-of-function genetics, not a direct

biochemical property of the CLV2 protein. The curation decision should keep these layers separate: reject the direct-binding MF, retain the complex-formation MF/CC, and tag the developmental BP annotations as genetic/loss-of-function in origin.

Curation Leads (require curator verification)

- **Candidate reference (refuting):** PMID:25754504 — snippet to verify: *"We showed that CLV2 and RPK2 exhibited no direct binding to the CLV3 peptide. Further analysis showed that the receptor kinase BAM1 directly binds the CLV3 peptide."*
- **Candidate reference (conflicting, must be addressed):** PMID:20626648 — snippet: *"We also find that CLV2 alone generates a membrane-localized CLE binding activity independent of CLV1."*
- **Candidate reference (reconciling principle):** PMID:20580598 — snippet: *"Arabinosylation of CLV3 is important for its biological activity and for high-affinity binding to its receptor, CLV1."*
- **Candidate reference (alternative supported MF):** PMID:19843317 — snippet: *"CLV2 can directly interact with CRN in the absence of CLV3."*
- **Candidate reference (architecture):** PMID:27229734 — snippet: *"CLV2, a leucine-rich repeat (LRR) receptor-like protein, and CRN, a protein kinase, have to interact to form a receptor-kinase complex. The kinase domain of CRN has been reported to be catalytically inactive."*
- **Suggested action change:** Replace/avoid any *peptide hormone binding* assertion with co-receptor framing (GO:0046982; GO:0043235); keep SAM/RAM BP annotations tagged as loss-of-function/genetic.
- **Suggested curator question:** Is the historical CLV2 binding annotation traceable to PMID:20626648 with a non-arabinosylated ligand? If so, flag as superseded by PMID:25754504.
- **Suggested experiment:** Quantitative direct binding (SPR/MST) on purified CLV2 ectodomain with native arabinosylated CLV3, CLV1 as positive control.

Summary Statement

The hypothesis that the CLV2 ectodomain directly binds CLV3/CLE peptide is **refuted** by the best available evidence. CLV2 is a kinase-less LRR receptor-like protein that functions as a non-binding co-receptor/scaffold via obligate heterodimerization with the CRN pseudokinase; direct CLV3 perception is carried out by the LRR receptor kinases CLV1 and BAM1 using the native arabinosylated ligand. GO:0017046 should not be asserted for CLV2.
